

AUTOMOBILE IDEATION CHALLENGE

Gather & Gain insight of data
effectively and automatedly

GROUP 4 - NeuroDrive

Using Data Scraping and AI for Car's Consumer Insight

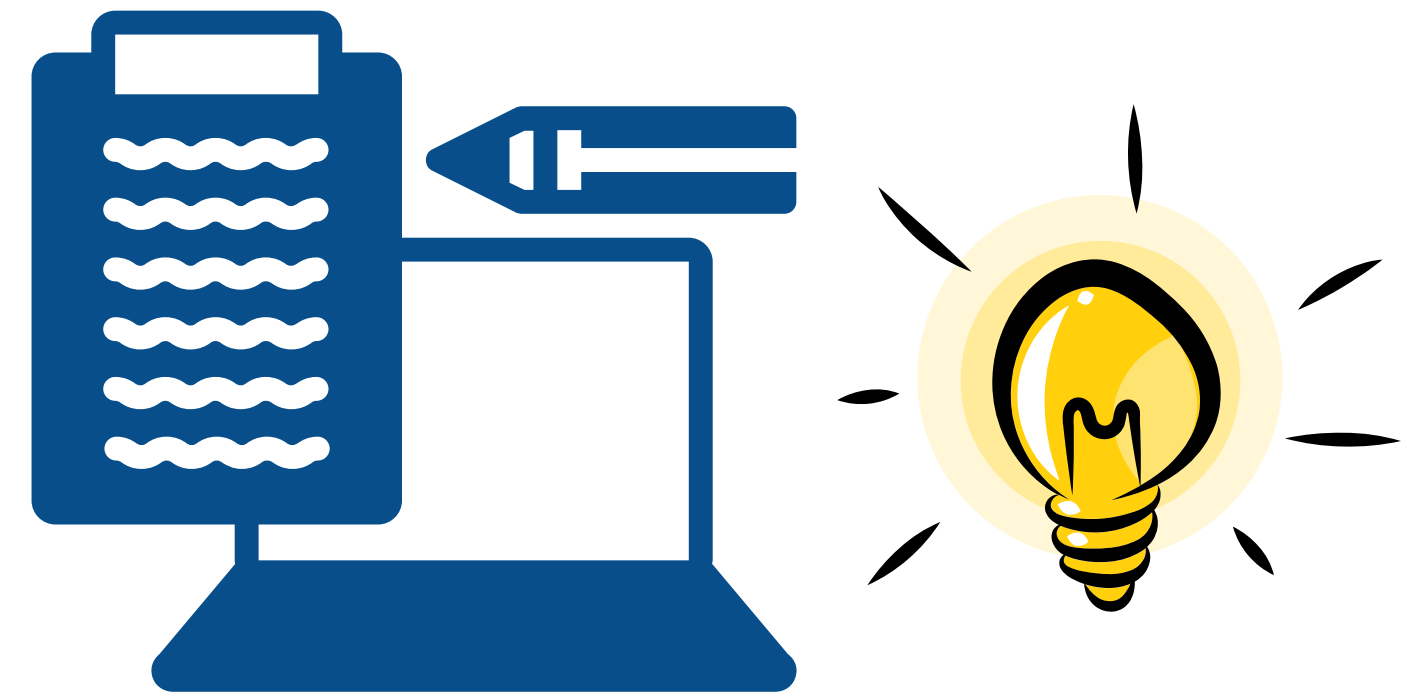
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Solution

Solution Stages

Expected Insight

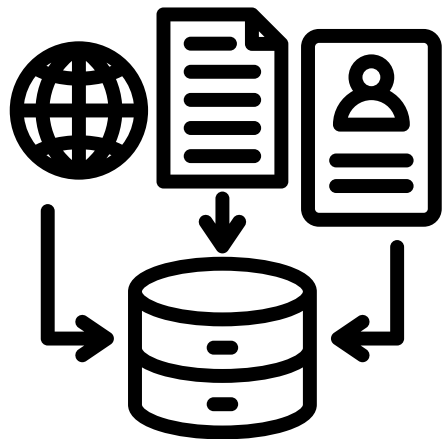
Unique value proposition



SOLUTION

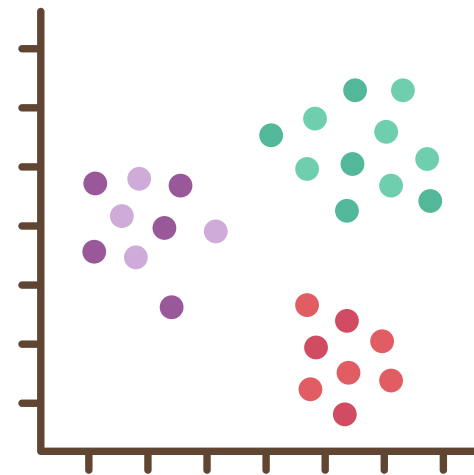
This system will aggregate data from multiple sources, correlate it using machine learning, and provide actionable insights through the use of proxy variable without relying on direct dealership cooperation or disrupting industry norms.

Stage 1



Data Gathering

Stage 2



Data Analysis

Stage 3



Insight

SOLUTION: PROOF OF CONCEPT

Proof of Concept



rely heavily on
scraped product listings



public professional data to build
predictive models of consumer behavior

By replacing surveys with public data, we simplify the data collection process, reduce costs, and maintain a manageable scope, all while preserving the accuracy and depth of insights

SOLUTION STAGES

STAGE 1: DATA SCRAPING

The system collects data from platforms like Khmer24, Facebook. These data points will be foundational for later analysis and modeling.



khmer24



facebook

DATA SCRAPING: KHMER24

14765 Result On 27 Sep, 2024

View: [Grid Icon] [List Icon] Sort By: Latest Ads ▾



Mercedes Benz C300 2015 Luxury 4Matic 🚗 សំដាប់មេឡាន ធានាជូនទឹកថ្នាំស៊ីនម៉ាជី

Mercedes Benz C300 2015 Luxury 4Matic 🚗 សំដាប់មេឡាន ធានាជូនទឹកថ្នាំស៊ីនម៉ាជី

Phnom Penh 57 minutes ago

\$33,500



Mercedes CLA45 AMG 2014 ឆ្នាំ

ឡានថ្មីមេឡាន សំដាប់មេឡាន ធានាជូនទឹកថ្នាំស៊ីនម៉ាជី

Phnom Penh 1 hours ago

\$30,000



ឡានឆ្នាំថ្មី

មកមើលឡានថ្មី ចាប់ពី ឡានថ្មីមេឡាន ធានាជូនទឹកថ្នាំស៊ីនម៉ាជី

Telegram 085999909

Phnom Penh 2 hours ago

\$69,900



Lexus Nx 200t F-sport for sale

Car in good condition, interest pls contact telegram 070 839860

Phnom Penh 5 hours ago

\$37,000 ~~\$37,800~~

Home / Cars and Vehicles / Cars for Sale / Mercedes-Benz / C300



Mercedes Benz C300 2015 Luxury 4Matic 🚗 សំដាប់មេឡាន ធានាជូនទឹកថ្នាំស៊ីនម៉ាជី

\$33,500

Ad ID : 11413338 Category : Cars for Sale Locations : Phnom Penh Posted : 27-Sep-24

Description

Car Makes:	Mercedes-Benz	Car Model:	C300	Year:	2015
Tax Type:	Tax Paper	Condition:	New	Body Type:	Sedan
Fuel:	Petrol	Transmission:	Auto	Color:	White

Auto

Last Active Yesterday

Show all ads from this customer

077935xxx

Click to show phone number

<https://www.khmer24.com/en/8889Auto>

Phnom Penh, ភ្នំពេញ, មូលដ្ឋាន, ទឹកល្អក់ទី ៣៨៨ 89 AUTO077 999 900<https://goo.gl/maps/V23JcH1marjp2X7g6>

Show on Google Map

Chat

Car Makes	Mercedes-Benz
Car Model	C300
Year	2015
Fuel	Patrol
Color	White
Transmission	Auto
Condition	New
Body Type	Sedan
Tax Type	Tax Paper
Location	Phnom Penh , ភ្នំពេញ, មូលដ្ឋាន, ទឹកល្អក់ទី
Listing Date	22/04/2000



318 friends • 43 mutual

[Add friend](#) [Message](#)

Posts **About** Friends Photos Videos Sports More

About
[Overview](#)
Work and education
Places lived
Contact and basic info
Family and relationships
Details About Chrun
Life events

No workplaces to show

Studied at Institute of Technology of Cambodia

Lives in **Phnom Penh**

No relationship info to show

Basic info

Male

Gender

28 December

Birth date

2003

Birth year

Relationship

No relationship info to show

Family members

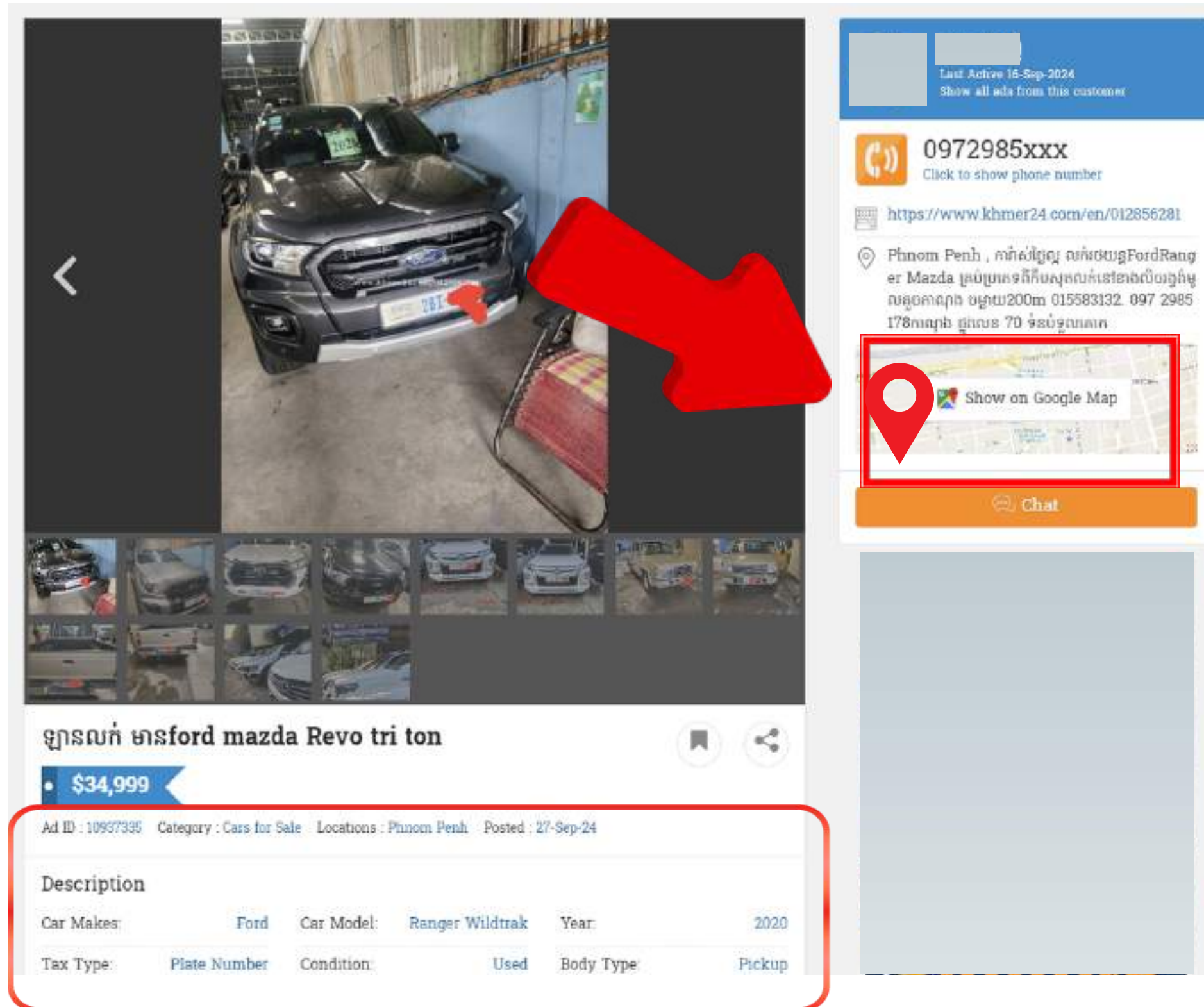
Sister

Background	ITC(Engineering)
Gender	Male
Age	21
Current town/city	Phnom Penh
Family members	1 sister
Relationship	None

PAGE 5

STAGE 2: DATA ANALYSIS

Geographic & Demographic Data from khmer24

khmer24

0972985xxx
Click to show phone number

<https://www.khmer24.com/en/012856281>

Phnom Penh, ភាគសំប៉ែត្នូ លក់រថយន្តFordRanger Mazda គ្រប់ប្រភេទទឹកបិទស្រទតលក់នៅខាងលិចភ្នំពេញ លក្ខណៈគុណៈ ម៉ូដែល200m 015583132. 097 2985 178ភាគរថ ថ្លៃលក់ 70 ទំនប់ទូលក់

Show on Google Map

Chat

ឡានលក់ មានford mazda Revo tri ton

\$34,999

Ad ID : 10937335 Category : Cars for Sale Locations : Phnom Penh Posted : 27-Sep-24

Description

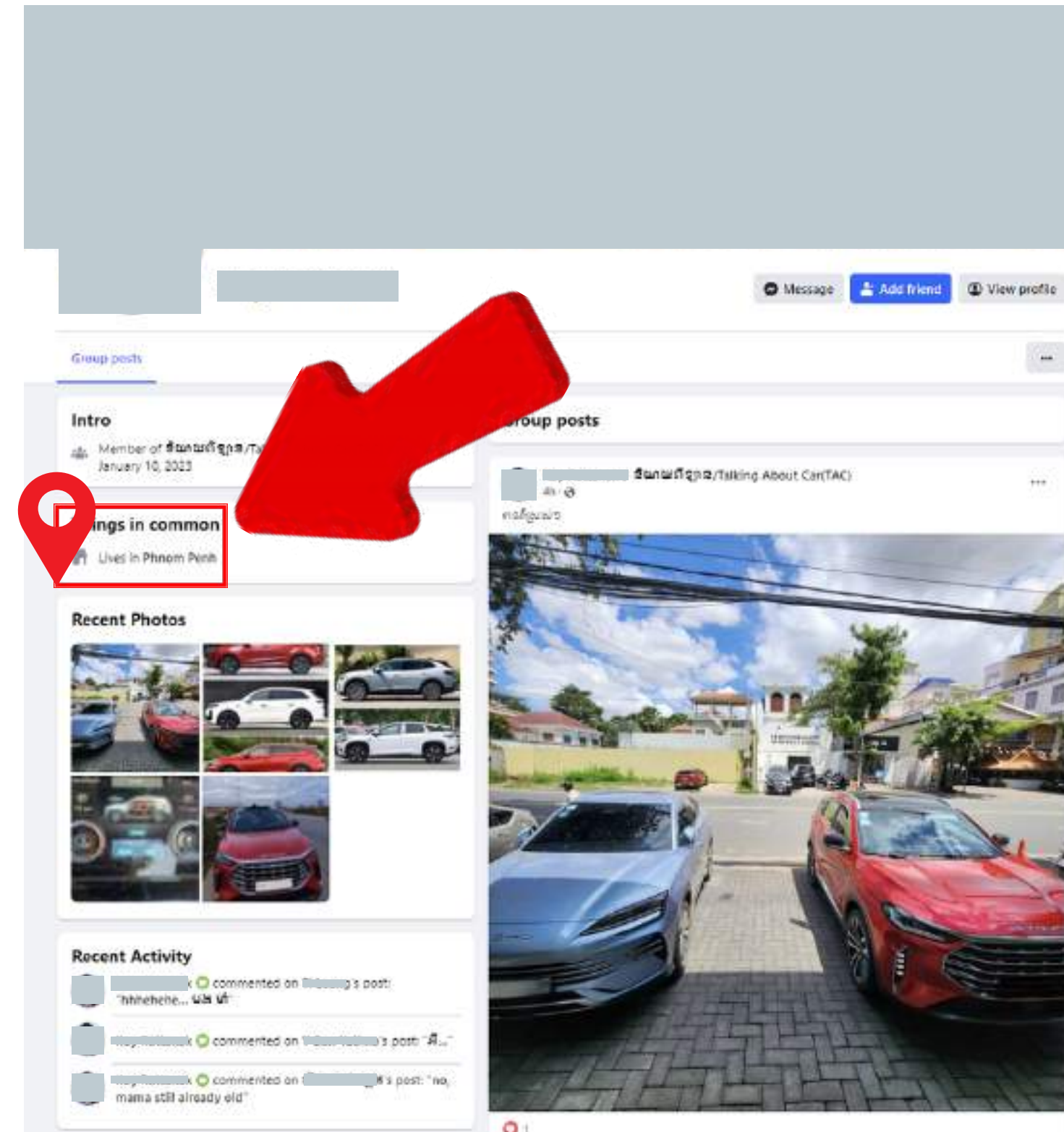
Car Makes:	Ford	Car Model:	Ranger Wildtrak	Year:	2020
Tax Type:	Plate Number	Condition:	Used	Body Type:	Pickup

We will **gain insight** and **classify** the car criteria in each **Location**

How do car criteria (types, colors, petrol and etc) vary by region on Khmer24, and how can this data guide location-specific inventory management for dealerships?

STAGE 2: DATA ANALYSIS

Geographic & Demographic Data from Facebook



- Beside gathering the contents post and comments to understand **consumer behavior**, **preference** or any **perspective** of any car brand
- Then we will able to classify group of car consumer's preference by **Location**



STAGE 2: DATA ANALYSIS

Raw Data from **khmer24** & facebook

post_id	post_date	author	content	car_make	car_model	car_year	car_price	car_color	car_condition	phone	location	image_urls
0	2024-09-27	Chan Dara	Mazda CX-5 ឆ្នាំ 2022 តម្លៃ 32,000 USD មាន...	Mazda	CX-5	2022	32,000 USD	អ	New	097654321	Battambang	[https://facebook.com/image/abc.jpg, https://...]
1	2024-09-25	Sok Khun	Ford Ranger ឆ្នាំ 2020 តម្លៃ 28,000 USD មានប្រ...	Ford	Ranger	2020	28,000 USD	លឿ	Used	0123456789	Phnom Penh	[https://facebook.com/image/def.jpg, https://...]
2	2024-09-24	Vannak Sok	Hyundai Elantra 2021 តម្លៃ 24,000 USD លក់នៅកំព...	Hyundai	Elantra	2021	24,000 USD	លឿ	New	071223344	Kampong Speu	[https://facebook.com/image/ghi.jpg, https://...]
3	2024-09-23	Mony Chan	Toyota Prius 2019 តម្លៃ 22,000 USD ដំឡើងប្រព័ន្ធ...	Toyota	Prius	2019	22,000 USD	អ	Used	09887766	Siem Reap	[https://facebook.com/image/jkl.jpg, https://...]
4	2024-09-22	Phirun Khem	Lexus RX 350 2018 តម្លៃ 40,000 USD ត្រូវបានកែលម្អ...	Lexus	RX 350	2018	40,000 USD	លឿ	Used	007987654	Phnom Penh	[https://facebook.com/image/mno.jpg, https://...]
5	2024-09-21	Srey Nang	Nissan Navara 2022 តម្លៃ 35,000 USD មានប្រព័ន្ធបញ្ជា...	Nissan	Navara	2022	35,000 USD	អ	New	009054321	Phnom Penh	[https://facebook.com/image/pqr.jpg, https://...]
6	2024-09-20	Dara Sovann	Chrysler Colorado 2019 តម្លៃ 30,000 USD មានប្រព័ន្ធបញ្ជា...	Chrysler	Colorado	2019	30,000 USD	លឿ	Used	005123456	Phnom Penh	[https://facebook.com/image/stu.jpg, https://...]
7	2024-09-19	Seyha Hun	Honda Civic 2020 តម្លៃ 27,000 USD មានប្រព័ន្ធបញ្ជា...	Honda	Civic	2020	27,000 USD	លឿ	New	090776655	Koh Kong	[https://facebook.com/image/vwx.jpg, https://...]

Raw Data from
khmer24



user_id	khmer_name	age	location	education
0	1	សុខ ចាន់	Phnom Penh	Bachelor's in Business Administration
1	2	ស្រី សុផា	Siem Reap	Master's in Public Health
2	3	ជី ធី	Battambang	Bachelor's in Information Technology
3	4	ជា វណ្ណា	Sihanoukville	Bachelor's in Education
4	5	វឿន ហ្វេត	Kampong Cham	Master's in Environmental Science
5	6	រ៉េត សុផាភី	Kampot	Bachelor's in Engineering
6	7	ជួន ពារា	Takeo	Diploma in Graphic Design
7	8	ខៀវ សុភា	Kampong Thom	Bachelor's in Marketing
8	9	ជា វិទ្យា	Svay Rieng	Master's in Finance
9	10	វ៉ែន ឆាយ	Kandal	Bachelor's in Agriculture

Raw Data from
facebook - User Data

post_id	post_date	author	content	comments	tags	location
0	1	Chan Dara	Mazda CX-5 ឆ្នាំ 2022 តម្លៃ 32,000 USD មាន...	[សូម ផ្គត់ផ្គង់ CX-5 ដូច គ្នា... តម្លៃទើបតែ...]	[Mazda, CX-5, Car Maintenance]	Battambang
1	2	Sok Khun	Ford Ranger ឆ្នាំ 2020 តម្លៃ 28,000 USD មានប្រ...	[ផ្ទុកប្រព័ន្ធបញ្ជាដោយស្វ័យប្រវត្តិ... មានប្រព័ន្ធបញ្ជាដោយស្វ័យប្រវត្តិ...]	[Ford, Ranger, Safety, Car Features]	Phnom Penh
2	3	Vannak Sok	Hyundai Elantra 2021 តម្លៃ 24,000 USD លក់នៅកំព...	[សូមបញ្ជាក់ទៅអ្នកទាំងអស់... ធ្វើការត្រួតពិនិត្យប្រចាំថ្ងៃ...]	[Hyundai, Elantra, Car Issues]	Kampong Speu
3	4	Mony Chan	Toyota Prius 2019 តម្លៃ 22,000 USD ដំឡើងប្រព័ន្ធបញ្ជា...	[ត្រូវបានកែលម្អប្រព័ន្ធបញ្ជាដោយស្វ័យប្រវត្តិ... ផ្ទុកប្រព័ន្ធបញ្ជាដោយស្វ័យប្រវត្តិ...]	[Toyota, Prius, Hybrid, Fuel Efficiency]	Siem Reap
4	5	Phirun Khem	Lexus RX 350 2018 តម្លៃ 40,000 USD ត្រូវបានកែលម្អ...	[ផ្ទុកប្រព័ន្ធបញ្ជាដោយស្វ័យប្រវត្តិ... ធ្វើការត្រួតពិនិត្យប្រចាំថ្ងៃ...]	[Lexus, RX 350, Smart System, Car Features]	Phnom Penh

Raw Data from
facebook - Content Post

Final Dataset from khmer24 & Facebook

A. Data Table from Khmer24 (Reselling Platform)



Car_Make_Model	Year	Price (USD)	Condition	Location (Province)	Mileage (Km)	Fuel_Type	Transmission_Type	Number_of_Views	Seller_Contact_Info
Toyota Camry	2016	35,000	Used	Phnom Penh	50,000	Petrol	Auto	2,000	0965364xxx
Honda CR-V	2018	22,000	Used	Siem Reap	30,000	Diesel	Manual	1,500	0977011xxx
Lexus NX	2015	43,500	Used	Phnom Penh	60,000	Petrol	Auto	2,500	0977222xxx
Ford Ranger	2012	15,000	Used	Battambang	80,000	Petrol	Manual	3,200	0995432xxx
Hyundai Tucson	2017	12,500	Used	Kampong Cham	45,000	Petrol	Auto	1,800	0881234xxx

B. Data Table from Facebook (Public Profiles)



User_ID	Location (Province)	Employment_Info (Job/Company)	Age	Interests_Groups	Public_Posts_Interest	Friends_Social_Connections
FB_001	Phnom Penh	Software Engineer at ABC Corp	34	Toyota Enthusiasts, Car Resellers	Posted about Toyota Camry	Connected with car dealers
FB_002	Siem Reap	Marketing Executive at XYZ Ltd	29	Hybrid Car Buyers, Eco Drivers	Posted about Honda CR-V	Connected with car enthusiasts
FB_003	Phnom Penh	Business Owner at Private Firm	45	Luxury Car Club, Lexus Owners	Discussing Lexus models	Connected with luxury car owners
FB_004	Battambang	Student at University of Battambang	22	Pickup Enthusiasts, Car Trading Group	Interested in off-road cars	Friends with off-road drivers
FB_005	Kampong Cham	Accountant at Accounting Firm	38	Hyundai Car Group, Diesel Fans	Liked a post about Hyundai Tucson	Connected with diesel car buyers

We will apply **AI algorithm** to transform these raw data through **Text Analysis**.

2

STAGE 2: DATA ANALYSIS

Objective of Dataset **after linking geographic and demographic by location**

To understand correlation of the group of people that consider owning a car in Cambodia context



Customer Segment



Product Development



Highlight the important criteria

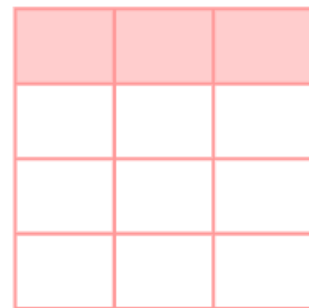


STEP 3: INSIGHTS

Step 1: Clustering (Persona Identification) segments

Persona Identification Segments

Facebook Dataset





Apply Clustering
Model



Luxury Vehicle Enthusiasts

- Location: Phnom Penh
- Age: 34-35
- Job: Doctor
- Interests: Active in luxury car clubs.

Eco-Conscious Young Professional

- Location: Battambang
- Age: 38-45
- Job: Electrician
- Interests: Engaged in pickup and budget-friendly car groups.

Young Enthusiasts

- Location: Phnom Penh
- Age: 22-29
- Job: Students or entry-level professionals
- Interests: Engaged with automotive enthusiast groups.

STEP 3: INSIGHTS

Step 2: Clustering the trending cars
base on location

Khmer24 Dataset





On-Trend

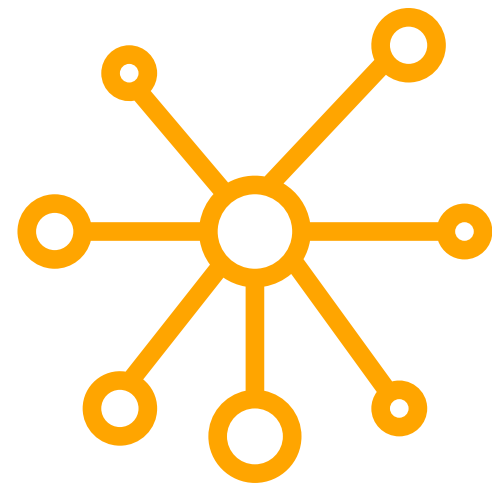
- List Date: 1 Week
- Status: Sold
- Car Specification:..
- Location: PP

Off-Trend

- List Date: 3 months
- Status: Sell
- Car Specification:..
- Location: PP

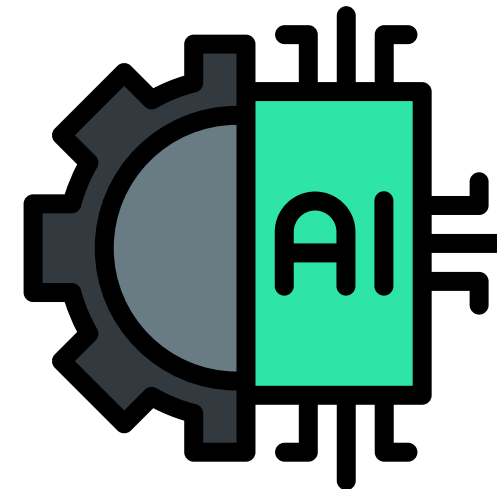
STEP 3: INSIGHTS

Step 3: Recommend Car Preference Using Machine Learning



Clustered Data of
Persona
Identification

Apply
→



AI

Result in
→



Recommend the
trending car into the
customer preference

EXPECTED INSIGHT

EXPECTED INSIGHT

Customer Persona

**35-44: Budget-Friendly Cars
(Toyota Corolla) - Affordability**

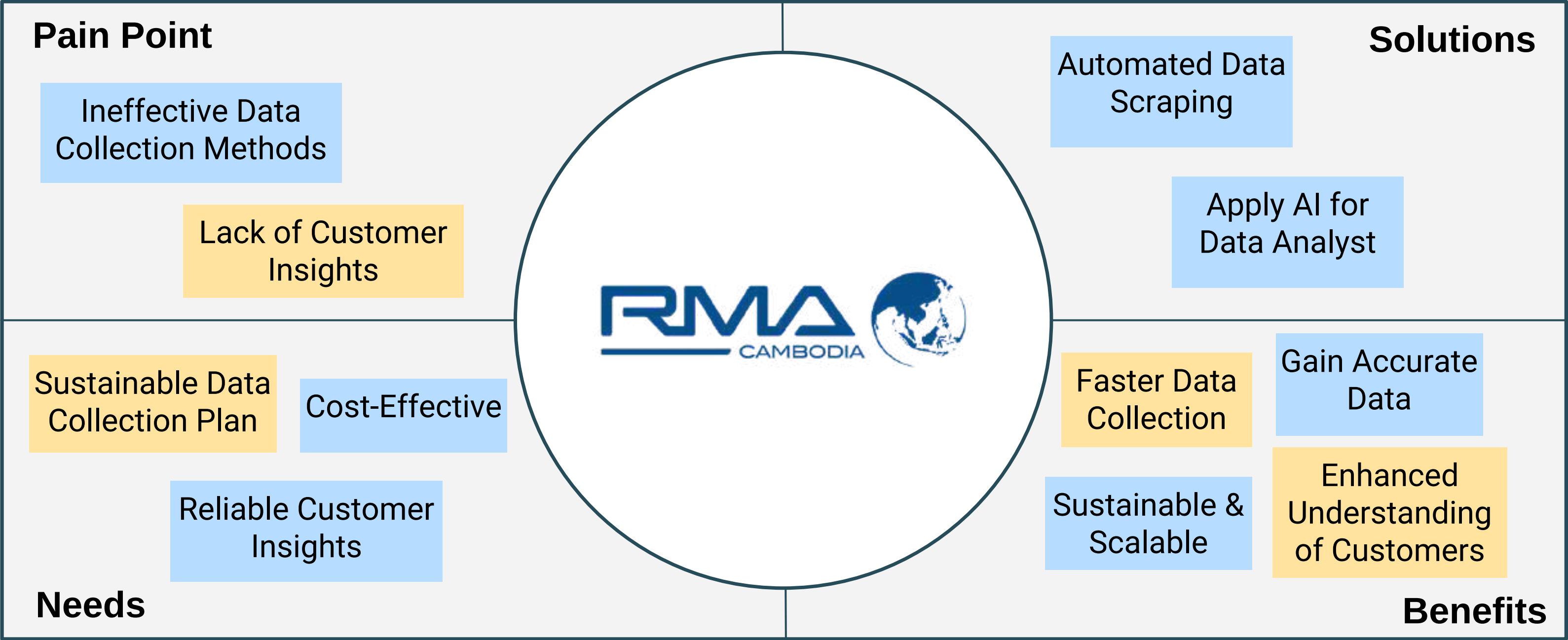
- Job: Mid-level professionals (e.g., engineers, managers).

Market Insight

- Toyota Camry: High sales in Phnom Penh - Luxury Demand
- Aligns with Customer interests - Marketing Opportunities

UNIQUE VALUE PROPOSITION

UNIQUE VALUE PROPOSITION



THANK YOU!

APPENDIX

Retail Industry (Walmart)

What they did:

- Used **demographic** data and **purchasing patterns** to predict **product demand by location**
 - In wealthier areas, they stocked higher-end products.
 - In rural areas, they focused on essential, low-cost items.

This is similar to how car types can be matched to customer demographics in different regions.



AFTER THE ANALYSIS, THE DATA WILL REVEAL CUSTOMER PERSONAS, PREFERENCES, AND BEHAVIORAL PATTERNS. THESE INSIGHTS WILL HELP RMA TAILOR ITS MARKETING STRATEGIES AND IMPROVE CUSTOMER ENGAGEMENT

- Data Transformation into Actionable Insights
- Customer Personas

2. Data Transformation into Actionable Insights

Example 1: Car Importation Strategy

Data Collected:

- **Khmer24:**
 - Location: Phnom Penh
 - Car Model: Toyota Camry (2018), Hybrid, \$25,000
 - Seller's Location: Phnom Penh
 - Mileage: 60,000 km
- **Facebook:**
 - Location: Phnom Penh, job title: Engineer at a tech firm, age: 35.
 - Interest: Member of "Hybrid Cars Cambodia" Facebook group.

Actionable Insight: Based on the above data, you could infer that in Phnom Penh, mid-range professionals (engineers) are interested in eco-friendly cars, particularly hybrid models like the Toyota Camry. If a pattern emerges where a significant portion of this demographic is part of "Hybrid Cars Cambodia" and engages with similar content, it signals an opportunity for dealerships to focus on **importing hybrid cars into Phnom Penh**, especially targeting middle-aged professionals.

Resulting Action: A car dealership or importer could adjust their inventory to focus on eco-friendly vehicles, such as hybrids or electric cars, specifically targeting marketing efforts to professionals working in tech-related fields in Phnom Penh.

OUR TEAM

**Finally, the team be at the
first place**

